adaptor to drive the relay and the Arduino (Board2).

Configuration of Yoda module

Arduino Nano boards (Board1 and Board2) are not required at the initial configuration stage for both transmitter (YM1) and receiver (YM2) circuits. Connect Yoda module with adaptor as shown in Fig. 3 for both transmitter and receiver.

Fig. 3: Yoda adaptor and its mounting on Yoda module

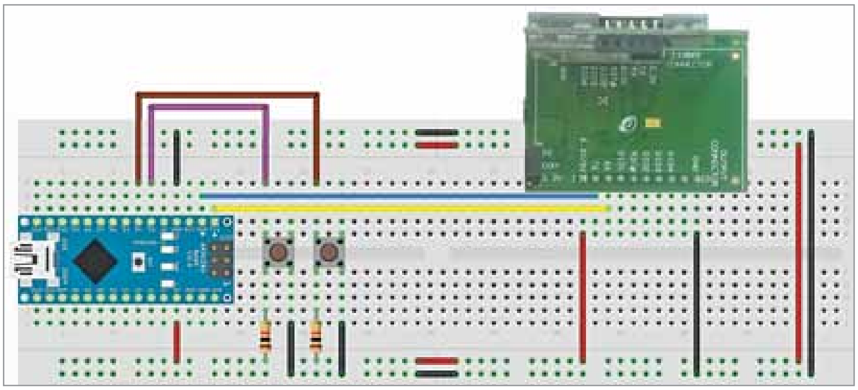

Fig. 4: Transmitter circuit wired on a breadboard

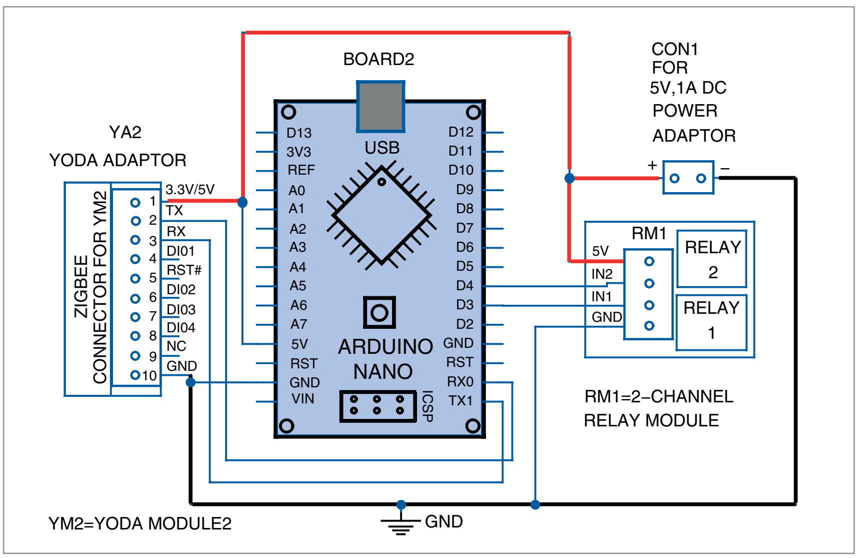

Fig. 5: Receiver circuit

You need serial communication software such as Docklight in your Windows PC. Docklight is a testing, analysis, and simulation tool for serial communication protocols. It is basically a serial terminal tool like TeraTerm and RealTerm. You can download Docklight from *https://docklight.de/downloads/*

To connect a Yoda adaptor to your PC, first you need to install MCP2221 USB driver (USB 2.0 to UART protocol converter) in your PC. The latest driver for MCP2200/MCP2221 can be downloaded from *https://www.microchip.com/wwwproducts/en/MCP2221*. Click on Documents option to get the exact driver.

Both YM1 and YM2 need to be configured properly in Docklight to start communication between the two modules. Each Yoda module (YM1 and YM2) should be configured with common channel and PAN ID values to interact/communicate between the two or more Yoda modules. The Yoda Command Set document describes Yoda module software configurations and the packet structure for sending data from one Yoda module to another.

You can test the two circuits with a computer by connecting YA1 to a USB port and YA2 to another USB port of the same laptop/PC.

Configuration procedures

Follow the steps below to configure a Yoda module:

1. Install Docklight v2.3 evaluation version in your PC/laptop. Open Docklight and click OK in Docklight License Registration window. Refer Yoda Zigbee Adaptor manual for step-by-step hints for hardware configuration.

2. Install MCP2200/MCP2221 driver in your system.

3. Connect Yoda module with its adaptor, as shown in Fig. 3. Then connect Yoda adaptor board to your PC/laptop using the micro USB cable. To check the communication between PC and Yoda module, first send 'Dump Connection' command. If you get response from Yoda module (YM1), PC to Yoda module connection is working. You can now test the other configuration commands also as described below.

4. Connect receiver YM2 to PC. Select the correct COM port in Docklight for both YA1 and YA2.

5. Configure channel and PAN-ID in both Yoda modules (YM1 and YM2) with the help of Write command by referring to the Yoda Command Set document. (During testing we used PAN-ID as 0001 and Channel as 11 for both transmitter and receiver.)

6. Execute Read command for confirming the configurations.

7. Restart/reset both Yoda module circuits.